



Electrifuell team

out of popular encouragement and personal dedication to revolutionising Indian electric mobility. The startup provides the needed inside infrastructure for battery manufacturers and EV OEMs and claims to reduce their market turnaround time.

“We are building a complete EV technology stack starting from the physical layer which will have BMS, VCU, onboard and offboard chargers, swapping station, and telematics module to the back-end SaaS infrastructure,” explains Ujjwal when questioned about their current venture. The 2017-born startup caters to battery manufacturers, fleet operators, and swapping station operators. He clears that Electrifuell has not yet entered into the domain of components like clusters, as they are very ergonomically and uniquely designed for every vehicle model.

Currently, the startup provides 16 hardware offerings under three verticals related to BMS, telematics, and charging. Talking about their “chemistry-agnostic” 16S Smart BMS designed especially for EVs, Ujjwal notes that it is certified for the latest

AIS 156 standard requirements and supports battery paralleling and hot swapping as well. Electrifuell boasts of providing customised features to battery manufacturers, like application-optimised hardware, onboard

data storage, firmware functions, and io options which are not provided by other BMS offerings in the market. As per Ujjwal, their products are priced in the range of ₹1,000-8,000, depending on the variant.

Ujjwal recalls that back when they started, China was a dominating player from where all the supply was being sourced. “We saw huge gaps in battery and EV product quality, functionalities, feature sets, and support in Indian conditions and we entered the domain to fill these multiple chasms,” he states as the founders’ motivation.

Interestingly, the four founders used to participate in various robotics contests, hackathons, etc and had worked together in high-pressure environments. Ujjwal, senior to the other three, even had a stint in an EMS company to know the working of the startup ecosystem deeply.

Soon after getting in touch with battery manufacturers, they got pseudo-incubated at their own college campus in Bahal, utilised the lab infrastructure there to get the hang of the tech and stack, and got

real traction there. Later in 2019, the startup got incubated at MeitY’s Electropreneur Park and grew so much that they sold BMS and grabbed their first boot-strapped revenue of ₹22.5 million in no time.

Close to hitting 5,000 orders a month, the company aims for five-digit orders by Q1 of the next financial year. It is currently handling a small chunk of the NCR market and plans to expand pan-India soon.

Having recently raised a whopping ₹18 million in funding from the Indian Angel Network, the startup claims to devote over half of its resources to its research and development wing. As per Ujjwal, they are evaluating multiple BMS, power electronics and semiconductor offerings from multiple sources. “We are working on an automotive array of BMS offerings, improved 4G LTE telematics versions, vehicle control units, and scalable charger platform. We are very speedily developing our ATM for batteries, i.e., swapping station platform that can be utilised by any swapping station operator.”

The CEO further conveys that Electrifuell provides internships and fresher “nerds” are welcome in the firm. The 26-people-strong team is looking to scale in South America, Africa, and Asia Pacific and partner with EV OEMs and auto ancillaries. “We dream of having one out of every 10 electric two-wheelers in India with Electrifuell’s technology in them,” Ujjwal said excitedly when asked about their next big milestone.

—Vaishali Yadav

3 UTILISING DATA FOR BETTER BATTERY SYSTEMS

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Lithium-ion batteries have transformed the energy space, but even as the nascent sector struggles to secure its footing in a world awakening to the effects